

HAMILTON–BOLTZMANN COTANGENT GEOMETRY FROM DETERMINISTIC HARD-SPHERE COLLISIONS

QIAN QI

ABSTRACT. The density–contact covariance is constructed first from exact finite-volume cumulants and is driven by an isonormal Gaussian measure on collision space. Its coefficient is the complete $\Delta p + \psi$, so pure contact and density–contact brackets are present. A perturbative nonautonomous energy theorem types the regular biased chart, while process tightness uses unconditional deterministic-interval cumulants with the required microscopic diagonal term and a multiscale grid, not an impossible full-state stopping-time estimate. The balanced gauge includes endpoints and transport, and the second epi-derivative is the closed Cameron–Martin form on $\text{Ran } \Sigma^{1/2}$.

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1. ROUND-FOURTEEN POSITIVE CLOSURE: THE JOINT COLLISION GAUSSIAN FORM

The Gaussian field is constructed first from finite-volume joint cumulants. It is not inferred from a formal Hessian of the action. Its collision driver acts on the complete coefficient $\Delta p + \psi$, so pure contact observables have nonzero variance. Tightness is proved from unconditional deterministic-interval

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estimates and a multiscale grid; no stopping-time estimate conditioned on the complete deterministic microstate is used.

1.1. The collision quotient and balanced source space. Let

$$\mathfrak{C}_T = [0, T] \times \mathbb{T}^3 \times \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{S}^2 / \sim, \quad (v, v_*, \omega) \sim (v_*, v, -\omega),$$

and also identify pre/post variables by the elastic involution when a test is declared symmetric. Let $\mathcal{Y}_\psi$ be the exponential Orlicz heart for the reference contact measure A_f . A density test p belongs to $\mathcal{Y}_p$ when its endpoint traces, transport derivative, and

$$\Delta_\omega p = p(v') + p(v'_*) - p(v) - p(v_*)$$

belong to the corresponding weighted spaces.

Define

$$\mathfrak{D}(p, \psi) = \Delta p + \psi.$$

The full finite-horizon balance coboundary generated by r is

$$\mathfrak{G}r = (r, -\Delta r, -r(0), r(T), -(\partial_t + v \cdot \nabla_x)r),$$

with all endpoint and transport components included.

Theorem 1.1 (Closed balanced gauge). *The weak balance operator from density/contact paths to endpoint, transport, and collision distributions has closed range on the short-time weighted Maxwellian chart. Its annihilator is exactly the closure of the full coboundaries $\mathfrak{G}r$ together with the five global conservation constraints. On a balanced pair the dual action depends only on $\mathfrak{D}(p, \psi)$.*

Proof. Solve the backward transport equation with terminal data on the torus and use the collision Dirichlet form on each velocity fiber. On the complement of the five invariants, the Maxwellian lower bound and the small bounded collision tilt give

$$\int |\Delta r|^2 dA_f + \|(\partial_t + v \cdot \nabla_x)r\|_{-1, w}^2 \geq c \|r\|_{\mathcal{Y}_p/\mathcal{I}}^2.$$

This is a finite-time observability estimate, not a long-time Livšic claim. The closed-range theorem gives the asserted range and annihilator. Substitution of the weak balance identity reduces every dual pair to the single collision coefficient $\Delta p + \psi$. $\square$

1.2. The regular biased chart. Let $q = e^{\Delta p + \psi}$, and let

$$dA_f = \frac{1}{2} f f_* ((v - v_*) \cdot \omega)_+ dt dx dv dv_* d\omega.$$

Lemma 1.2 (Perturbative nonautonomous collision estimate). *On a sufficiently small bounded source chart produced by B2, the density satisfies*

$$cM(v) \leq f(t, x, v) \leq CM(v), \quad 0 < q_- \leq q, q^* \leq q_+,$$

where q^* is the pre/post transform. The symmetric part of the linearized collision operator has Dirichlet form comparable with

$$\frac{1}{4} \int (q + q^*) f f_* (h' + h'_* - h - h_*)^2 ((v - v_*) \cdot \omega)_+,$$

and is coercive modulo the five velocity collision invariants. The nonautonomous transport-collision equation is well posed on $[0, T_0]$ and obeys a weighted finite-time energy estimate.

Proof. B2 constructs the source-biased solution by Picard iteration from a Maxwellian-comparable datum. Small time and a bounded source preserve the two-sided Maxwellian comparison and the positive bounds on q, q^* . Average the collision bilinear form with its pre/post adjoint. The displayed nonnegative square is the leading term; all coefficient differences from the equilibrium form are relatively bounded by the source radius. The classical Maxwellian collision gap is therefore stable under this perturbation.

For spatially dependent coefficients, do not diagonalize by Fourier transform. Commute the equation with the hypocoercive multiplier $\nabla_x(1 - \Delta_x)^{-1} \nabla_v$, estimate coefficient commutators by their bounded chart norms, and integrate the resulting differential inequality. The transport part is skew after the commutator correction, the microscopic part is controlled by the collision gap, and the five local moments satisfy the exact finite hyperbolic balance coupled to the controlled higher moments. Gronwall gives the finite-time estimate. $\square$

1.3. Covariance from microscopic cumulants. Under a regular biased microscopic law, define

$$Z_\varepsilon(p, \psi) = \sqrt{\mu_\varepsilon} [\langle p, \pi^\varepsilon - f \rangle + \langle \psi, \Gamma^\varepsilon - \Gamma \rangle].$$

B2 gives normal convergence of the joint source pressure on a complex neighborhood.

Theorem 1.3 (Joint covariance kernel). *The finite-dimensional laws of Z_ε converge to a centered Gaussian field Z . Its dynamic martingale bracket is*

$$\begin{aligned} & \langle M(p, \psi), M(\tilde{p}, \tilde{\psi}) \rangle_t \\ &= \int_0^t \int q(\Delta p + \psi)(\Delta \tilde{p} + \tilde{\psi}) dA_f, \end{aligned}$$

and its initial covariance is the B1 microcanonical Schur complement. Equivalently, there is an isonormal Gaussian random measure $\mathbb{W}$ on $L^2(qA_f)$ such that the direct collision noise is

$$M_t(p, \psi) = \int_{[0, t] \times \mathfrak{c}} (\Delta p + \psi) \sqrt{q} d\mathbb{W}.$$

In particular a pure contact test has variance $\int q\psi^2 dA_f$.

Proof. For $k \geq 3$, the k -th centered cumulant of the $\sqrt{\mu_\varepsilon}$ -scaled field is $O(\mu_\varepsilon^{1-k/2})$, by the normal B2 connected expansion. The second cumulant

converges by two source derivatives. Differentiating the collision exponential $e^{\Delta p + \psi}$ twice gives exactly the displayed product, including contact–contact and density–contact terms. B1 supplies the conditioned initial block. The limiting positive bilinear form defines the isonormal measure and the Gaussian finite-dimensional laws. $\square$

The linear density fluctuation is the unique solution of the nonautonomous linearized biased Boltzmann equation driven by this collision measure. Lemma 1.2 gives its continuous solution operator.

1.4. Deterministic-interval tightness. Let $\mathcal{F}_t^{\text{obs}}$ be the filtration generated by the empirical density/contact fields, not the complete microscopic state. We do not claim a Poisson conditional intensity in this filtration.

Lemma 1.4 (Localized cumulant moments). *For every smooth compact family of tests and every deterministic interval I of length h ,*

$$\mathbb{E}|Z_\varepsilon(I)|^{2m} \leq C_m \left(h^m + \sum_{\ell=1}^{m-1} \mu_\varepsilon^{-\ell} h^{m-\ell} \right).$$

In particular

$$\mathbb{E}|Z_\varepsilon(I)|^4 \leq C \left(h^2 + \frac{h}{\mu_\varepsilon} \right).$$

The estimate is unconditional and remains valid jointly for density and contact increments.

Proof. Insert time-supported sources into the B2 connected expansion. A connected cumulant with j distinct collision times has an L^1 density on I^j , hence contributes $O(h^j)$. Coincident contact insertions produce diagonal contractions and one lost factor of μ_ε . Summing partitions of $2m$ insertions gives the displayed powers. This is the correct microscopic diagonal term; no estimate conditioned just before a deterministically known collision is asserted. $\square$

Theorem 1.5 (Process-level joint Gaussian limit). *The density/contact fluctuation pair is C -tight in the weighted distribution path space and converges to the Gaussian solution determined by Theorem 1.3.*

Proof. Take a dyadic grid with mesh δ . Markov’s inequality and Lemma 1.4 with $m \geq 3$ give

$$\mathbb{P} \left(\max_{I \in \mathcal{D}_\delta} |Z_\varepsilon(I)| > \eta \right) \leq C_\eta (\delta^{m-1} + \mu_\varepsilon^{-1} \delta^{m-2} + \dots).$$

Sum over dyadic levels down to μ_ε^{-2} . Below that scale, the maximal jump is $O(\mu_\varepsilon^{-1/2})$, while the B2 contact-count exponential moment controls the probability of multiple jumps in one finest cell. Letting first $\varepsilon \rightarrow 0$ and then $\delta \rightarrow 0$ proves the modulus estimate. Mitoma’s criterion and weighted compact containment give C -tightness. Finite-dimensional convergence and uniqueness of the linear Gaussian equation identify the limit. $\square$

1.5. The Cameron–Martin form and the action tangent. Let Σ be the covariance operator on the quotient dual. Define the closed extended form

$$q(z) = \begin{cases} \|\Sigma^{-1/2}z\|^2, & z \in \text{Ran } \Sigma^{1/2}, \\ +\infty, & \text{otherwise.} \end{cases}$$

Theorem 1.6 (Second epi-derivative). *At every regular exposed path of the B2 good rate,*

$$d_e^2 I(z) = q(z).$$

The dual kernel of Σ is the closed full balance-coboundary space of Theorem 1.1; after quotienting it, q has no nonzero primal null vector.

Proof. On every finite family of sources, B2 convex duality and the analytic pressure expansion give the inverse of the finite covariance matrix. Projective monotonicity follows because enlarging the source family takes a supremum of finite quadratic duals. The supremum is precisely the closed form $\|\Sigma^{-1/2}z\|^2$ on $\text{Ran } \Sigma^{1/2}$. B2’s full LDP supplies the finite-rate recovery needed for Mosco limsup. The finite-horizon observability theorem identifies the dual kernel with exact balance coboundaries, rather than invoking a long-time Livšic theorem. $\square$

Theorem 1.7 (Round-fourteen B3 closure). *The hard-sphere density/contact fluctuations have a correctly typed joint collision-space Gaussian driver, unconditional deterministic-interval process tightness, and the Cameron–Martin action tangent on $\text{Ran } \Sigma^{1/2}$.*

Proof. Combine Theorems 1.3, 1.5, and 1.6, with Theorem 1.1. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-fourteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: qiqian@pku.edu.cn